

Energy Balance & Basal Metabolic Rate

Fundamentals of Human Energy Balance

- Energy balance represents the dynamic relationship between caloric intake from food and total daily energy expenditure for body functions.
- Maintaining a neutral energy balance means caloric intake perfectly matches expenditure, resulting in weight stability over a long period.
- Positive or negative energy states drive physiological changes, forcing the body to either store surplus nutrients or mobilize tissue reserves.

Defining Basal Metabolic Rate (BMR)

- Basal Metabolic Rate (BMR) is the absolute minimum caloric baseline required to sustain vital autonomic life functions while completely at rest.
- These involuntary processes include respiration, blood circulation, cellular synthesis, and maintaining basic neurological organ functions.
- BMR accounts for approximately 60% to 70% of a typical human's total daily energy expenditure before adding physical activity.

Primary Variables Influencing BMR

- Muscle tissue is highly metabolically active, meaning individuals with greater lean body mass inherently possess a significantly higher *BMR*.
- As chronological age increases, *BMR* typically decreases due to gradual sarcopenia and natural age-related shifts in internal hormone profiles.
- Biological sex affects baseline energy requirements primarily through divergent patterns of average muscle mass and circulating testosterone.

Practical BMR Calculation Methodology

- The Mifflin-St Jeor equation is the universally preferred standard formula due to its high predictive accuracy for modern diverse populations.
- Men calculate their baseline resting calorie estimate using:

$$BMR_{men} = (10 \times w) + (6.25 \times h) - (5 \times a) + 5$$

- Women calculate establishing their specific minimal energy needs using:

$$BMR_{women} = (10 \times w) + (6.25 \times h) - (5 \times a) - 161$$

- For these mathematical formulas, the variables represent: w for weight in kg, h for height in cm, and a for age in years.